Problem sheet 2 for Rigid Analytic Geometry Summer Term 2026

Let K be a field with an absolute value $|\cdot|_K$ satisfying the conditions from sheet 1. By a *normed K-vector space* we understand a K-vector space V with a map $V \xrightarrow{\|\cdot|V\|} [0,\infty)_{\mathbb{R}}$ such that the following holds for $x, y \in V$ and $\lambda \in K$:

- $\|x+y\,|V\| \leq \max(\|x\,|V\|, \|y\,|V\|)$
- $\|\lambda x\,|V\| = |\lambda|_K \|x\,|V\|$
- $\|x\,|V\| = 0$ implies $x = 0$.

In other words, the conditions are the same as for K-Banach spaces but with the completeness condition dropped. Under the above assumptions $\|\cdot\,|V\|$ is called a K-norm. We say that two K-norms $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent if there is $C \in [1,\infty)_{\mathbb{R}}$ such that

$$\|v\|_1 \leq C\,\|v\|_2 \leq C^2\,\|v\|_1$$

holds for all $v \in V$.

Let

$$\|(x_i)_{i=1}^m\,|K^m\| = \max_{1\leq i\leq m} |x_i|_K\,.$$

Obviously this turns K^m into a K-Banach space.

Problem 1 (1 point)**.** *If $\|\cdot\|_1$ is a K-norm on K^m, show that there is a real number C such that*

$$\|x\|_1 \leq C\,\|x\,|K^m\|$$

holds for all $x \in K^m$.

It is considerably harder to show that the opposite type of inequality also holds:

Problem 2 (4 points)**.** *If $\|\cdot\|_1$ is a K-norm on K^m, show that there is a real number C such that*

$$\|x\,|K^m\| \leq C\,\|x\|_1$$

holds for all $x \in K^m$.

Remark 1.

- *Thus, two K-norms on a finite dimensional K-vector space are equivalent.*
- *In classical functional analysis over $K = \mathbb{R}$ or $K = \mathbb{C}$ the anologous result is usually shown by considering the function*

  $$x \to \|x\|_1$$

  on $\left\{x \in K^m \;\middle|\; \|x\,|K^m\| = 1\right\}$. By an appropriate version of the first problem, it is continuous hence has minimum and

maximum. In our situation this will only work if K is locally compact. This holds for $\mathbb{Q}_p$ but not for other interesting fields like $\mathbb{C}_p$, the completion of the algebraic closure of $\mathbb{Q}_p$. Hence solutions which use compactness arguments are likely to work only under additional assumptions and are thus unlikely to be given the full number of points.

Problem 3 (2 points)**.** *Show that the assertion from the previous problem will no longer hold if the completeness assumption on $|\cdot|_K$ is dropped.*

From now on, K will again be assumed to be complete. We assume that L/K is a finite field extension.

Problem 4 (2 points)**.** *For a K-norm $\|\cdot\|$ on L, show that there is a positive real number C such that $\|xy\| \leq C\,\|x\|\,\|y\|$.*

The following was not shown in the lecture on Monday, April 20 as it is relatively obvious and one learns more by showing it by oneself.

Problem 5 (2 points)**.** *Let A be a ring, $\mathfrak{U}$ an ultrafilter on* Spec$A$ *and*

$$\mathfrak{g}_{\mathfrak{U}} = \Big\{ x \in A \;\Big|\; \big\{ \mathfrak{p} \in \mathrm{Spec}A \;\big|\; x \in \mathfrak{p} \big\} \in \mathfrak{U} \Big\}.$$

If $x \in \mathfrak{g}_{\mathfrak{U}}$ and $a \in A$, show that $ax \in \mathfrak{g}_{\mathfrak{U}}$.

Together with the considerations carried out in the lecture this shows that $\mathfrak{g}_{\mathfrak{U}}$ is indeed an element of Spec$A$.

It is worthwhile to work out on one's own a complete example of the use of generic limits of ultrafilters to show that a certain topological space is spectral. For the valuation spectrum of a ring this must be done in the lecture, but the example of the real spectrum is suitable and in fact a bit easier.

A *cone* in a ring A is a subset $C \subseteq A$ containing all squares in A and such that $x, y \in C$ implies $x + y \in C$ and $xy \in C$. For instance, if A is a ring of $\mathbb{R}$-valued functions on a set X then the subset of functions whose values on a given subset $M \subseteq X$ are ≥ 0 is a cone. We say that a cone is *proper* if $-1 \notin C$ and a *prime cone* if it is proper and $a, b \in A$, $ab \in C$ and $a \notin C$ implies $-b \in C$ (motivation: if $x \in X$ and $(ab)(x) \geq 0$ but $a(x) < 0$ then $b(x) < 0$).

Let Sper$A$ be the set of prime cones $\mathfrak{P} \subseteq A$. For $a \in A$, let

$$\mathcal{P}(a) = \big\{ \mathfrak{P} \in \mathrm{Sper}A \;\big|\; -a \notin \mathfrak{P} \big\}.$$

For $(a_i)_{i=1}^m \in A^m$, let

$$\mathcal{P}(a_1, \ldots, a_m) = \bigcap_{i=1}^{m} \mathcal{P}(a_i)$$

which in the case $m = 0$ must be taken to be $\mathrm{Sper}R$. Let

$$\mathfrak{B}_A = \left\{ \mathcal{P}(a_1, \ldots, a_m) \;\middle|\; m \in \mathbb{N} \text{ and } (a_i)_{i=1}^m \in A^m \right\}.$$

This is closed under finite intersections in $\mathrm{Sper}A$ hence can be taken as the base of a topology on $\mathrm{Sper}A$.

Problem 6 (8 points)**.** *Show that* $\mathrm{Sper}A$ *is a spectral space and that the elements of* $\mathfrak{B}_A$ *are quasi-compact.*

Let X be an infinite set and $\eta \in X$.

Problem 7 (2 points)**.** *Consider the topology on* X *where a subset* U *is open if* $U = \emptyset$ *or* $\eta \in U$ *and* $X \setminus U$ *is finite. Show that* X *is spectral!*

Spectralness is fairly easy to show in this case as all open subsets are quasicompact.

Problem 8 (2 points)**.** *For* X *from the previous problem, describe* X_{con} *by giving a neighbourhood base of each* $x \in X$*!*

Three of the 23 points from this sheet are bonus points which do not count in the calculation of the 50%-threshold for passing the exercise module.

Solutions should be e-mailed to your tutor s45qhou at uni hyphen bonn dot de before Monday May 4.